

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 18:30

File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	llama3.2

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MINOR REVISION

Weighted Score: 3.7 / 5.0
Confidence: 74%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	3.5	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.0	Acceptable	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 3.5 / 5.0 | Confidence: 80%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Weaknesses:

- [TITLE] could be more concise

Major Comments:

- [TITLE] could be more concise

Recommendations:

- [TITLE] Revise the title to make it more concise while maintaining its informativeness
- [ABSTRACT] Remove references and undefined abbreviations from the abstract

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 100%

Strengths:

- The manuscript presents a clear research problem and objectives.
- The structure flow is logical, with each section building upon the previous one.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Major Comments:

- The logical flow is generally good, but the sequence could be improved by explicitly stating the problem and then introducing the model.

Recommendations:

- Consider adding a section on equation dimensionality to ensure reproducibility.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 3.5 / 5.0 | Confidence: 60%

Strengths:

- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Weaknesses:

- Bootstrap confidence intervals detected but not justified with sufficient resamples
- Post-hoc corrections (Holm) applied correctly but not explicitly stated in the manuscript

Major Comments:

- The use of Kruskal-Wallis test for comparison is appropriate given the non-normal distribution of data.

- The application of post-hoc corrections (Holm) is correct, but it would be beneficial to explicitly state this in the manuscript.

Recommendations:

- Justify the use of bootstrap confidence intervals with sufficient resamples.
- Explicitly state the application of post-hoc corrections (Holm) in the manuscript.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the problem.

Weaknesses:

- Some results, such as Pareto analysis and hypervolume/spread indicators, are not explicitly tied to figures or tables.
- Limitations of results could be acknowledged more clearly.

Major Comments:

- The manuscript provides a clear overview of the proposed EANN framework and its comparison with GA and PSO baseline methods.

Recommendations:

- Include Pareto analysis and hypervolume/spread indicators in figures or tables for better clarity.
- Provide more explicit discussion on limitations of results to strengthen the manuscript.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.2 / 5.0** | Confidence: 80%

Weaknesses:

- The discussion could benefit from a more in-depth comparison with prior work, beyond just citing existing literature.

Major Comments:

- The conclusions are directly derived from the results and answer the original research question.

- The experimental validation is well-conducted, but it would be beneficial to discuss the limitations of the study more explicitly.

Recommendations:

- Consider adding a discussion on the theoretical implications of the proposed EANN framework.
 - Provide more detail on how the computational cost affects the practical utility of the proposed paradigm.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
 - Ensure technical terms are defined on first use and used consistently across sections.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Weaknesses:

- The references section could benefit from more recent publications (after 2021) to demonstrate the current state-of-the-art in the field.
- Some references appear to be incomplete or truncated, which may affect the accuracy of the citations.

Major Comments:

- While the manuscript provides a good overview of the current research landscape, it would benefit from more recent and seminal works to establish its originality and novelty.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Include at least one recent publication (after 2021) in the references section to demonstrate the current state-of-the-art in the field.
 - Verify the completeness and accuracy of all reference citations.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] could be more concise
- [2] The logical flow is generally good, but the sequence could be improved by explicitly stating the problem and then introducing the model.
- [3] The use of Kruskal-Wallis test for comparison is appropriate given the non-normal distribution of data.
- [4] The application of post-hoc corrections (Holm) is correct, but it would be beneficial to explicitly state this in the manuscript.
- [5] The manuscript provides a clear overview of the proposed EANN framework and its comparison with GA and PSO baseline methods.
- [6] The conclusions are directly derived from the results and answer the original research question.
- [7] The experimental validation is well-conducted, but it would be beneficial to discuss the limitations of the study more explicitly.
- [8] While the manuscript provides a good overview of the current research landscape, it would benefit from more recent and seminal works to establish its originality and novelty.
- [9] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [2] The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.
- [3] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. [TITLE] Revise the title to make it more concise while maintaining its informativeness
2. [ABSTRACT] Remove references and undefined abbreviations from the abstract
3. Consider adding a section on equation dimensionality to ensure reproducibility.
4. Justify the use of bootstrap confidence intervals with sufficient resamples.
5. Explicitly state the application of post-hoc corrections (Holm) in the manuscript.
6. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
7. Include Pareto analysis and hypervolume/spread indicators in figures or tables for better clarity.
8. Provide more explicit discussion on limitations of results to strengthen the manuscript.
9. Consider adding a discussion on the theoretical implications of the proposed EANN framework.
10. Provide more detail on how the computational cost affects the practical utility of the proposed paradigm.
11. Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
12. Ensure technical terms are defined on first use and used consistently across sections.

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